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AI-Based Digital Lifestyle and Wellbeing Analysis Among Young Adults

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Abstract

Mental health problems may develop gradually, making early changes in emotional wellbeing difficult to recognize. This paper presents SignalWell, a research-focused digital wellbeing platform that combines daily self-reported check-ins with digital activity measures to study changes in behaviour over time. The prototype uses a Next.js frontend, a FastAPI backend, and Supabase/PostgreSQL storage with Row-Level Security (RLS). It compares current measurements with an individual's recent personal baseline and verifies the availability of registered machine-learning artifacts before inference. Evaluation was conducted using simulated data rather than clinical or participant data. In the simulated risk-indicator task, gradient boosting achieved a mean absolute error (MAE) of 0.58, root mean squared error (RMSE) of 0.79, and coefficient of determination ($\boldsymbol{R^2}$) of 0.85. Security tests reported that unauthorized cross-account requests were blocked in the tested scenarios. A separate simulated alert comparison showed that personalized-baseline alerts detected more simulated events and reduced average detection delay, while exhibiting lower precision and a higher false-alert rate than fixed thresholds. These findings are preliminary and do not establish clinical validity. SignalWell is intended to support awareness of changes in digital habits and self-reported wellbeing, not to diagnose or replace professional mental-health assessment.

Keywords: Digital wellbeing, mental health monitoring, machine learning, digital phenotyping, personal baseline, privacy, Row-Level Security

1 Introduction

Mental health problems can develop gradually, and early changes in emotional wellbeing may be difficult to recognize. Traditional tracking often relies on self-reports and occasional assessments, which may not capture day-to-day variation. Digital devices create opportunities to observe routines over time, but translating activity signals into useful insights raises challenges related to privacy, security, individual differences, and ethical use.

This study introduces SignalWell, a cloud-oriented digital wellbeing platform that combines daily self-reported check-ins with digital activity measures. Rather than making a medical diagnosis, the platform examines changes relative to a user's own historical patterns, referred to as a personal baseline. The system uses a Next.js frontend, a FastAPI backend, and Supabase with PostgreSQL for data storage and authentication support. Database Row-Level Security (RLS) policies are intended to restrict access to user-specific records, while server-side model-registry checks gate prediction requests when required artifacts are unavailable.

The work investigates the following research questions:

- RQ1: Can basic digital activity tracking and daily mood check-ins indicate shifts in a user's wellbeing?
- RQ2: Does comparison with a personal baseline identify simulated risk events differently from fixed population-level thresholds?
- RQ3: Which features contribute most to the evaluated model's predictions?
- RQ4: How can separation of frontend, backend, and storage layers support privacy and restrict unauthorized access to behavioural records?

The hypotheses were that (H1) rolling personal baselines would identify simulated behavioural shifts earlier than fixed limits; (H2) database-level RLS would block cross-account access in the tested scenarios; and (H3) model-registry checks would prevent inference when required model artifacts were missing or invalid. The evaluation is limited to the simulated data and test scenarios described below.

2 Related Work

Digital phenotyping and passive sensing have been explored as ways to study behaviour and mental-health-related patterns using mobile and wearable technologies. Prior work also highlights the importance of temporal variation, data quality, privacy, and ethical oversight in digital mental health research. Machine-learning approaches can analyze behavioural and self-reported features, but results depend on the dataset, the target definition, and evaluation design. Model explanation methods such as SHAP can help characterize which features influence model outputs, although feature importance does not establish causality.

SignalWell combines personal-baseline comparison, database-level access controls, and runtime model-artifact verification in a single research prototype. The present experiments use simulated records; therefore, the findings should be interpreted as technical prototype evaluation rather than evidence of real-world or clinical performance.

Table 1 Selected related work as summarized in the original manuscript. The cited sources and their bibliographic details should be verified before submission.

Study	Contribution	Gap addressed by SignalWell
Mohr et al.	Passive sensing for depressive-symptom tracking	Separation of API gateway and runtime artifact validation
Torous et al.	Digital phenotyping frameworks mapping behaviours	Personal-baseline comparison rather than only static population metrics
Wahle et al.	Machine-learning pipelines using device interactions	Database-level RLS boundaries in the proposed platform

3 Proposed Methodology

SignalWell is a non-clinical research prototype designed to study behavioural patterns over time. It combines daily user-reported logs and broad digital activity measures to calculate a risk indicator. The system is designed to avoid collecting page contents, messages, or keystrokes and instead uses activity summaries.

3.1 System Architecture

The architecture has three principal layers: client interface, API/backend, and data storage. The frontend uses Next.js, React, TypeScript, and Tailwind CSS. It provides dashboards, consent controls, and visualizations. FastAPI validates requests, processes behavioural information, maintains personal-baseline calculations, and coordinates machine-learning inference. Supabase/PostgreSQL stores application data and applies RLS policies to user-specific records.

3.2 Data Fields and Metric Representation

The manuscript describes seven daily measures: sleep duration, movement, screen time, mood, stress, anxiety, and general wellbeing. Sleep duration and screen time are continuous measures; mood, stress, and anxiety are self-reported categorical or ordinal measures. The platform produces a risk indicator on a scale from 0 to 10, where a higher value represents a greater deviation from the user's usual pattern. The precise target-generation procedure and label semantics should be documented alongside the dataset before publication.

3.3 Personal-Baseline Model

For a feature vector $\mathbf{x}_t$ on day t, the deviation from the mean of the preceding seven valid days is defined as

$$\Delta \mathbf{x}_t = \mathbf{x}_t - \boldsymbol{\mu}_{t-1:t-7}, \tag{1}$$

where $\boldsymbol{\mu}_{t-1:t-7}$ is the feature-wise mean over the preceding seven valid days. The system requires seven days of history to establish the initial baseline. This is a within-user comparison and should not be interpreted as a clinical threshold.

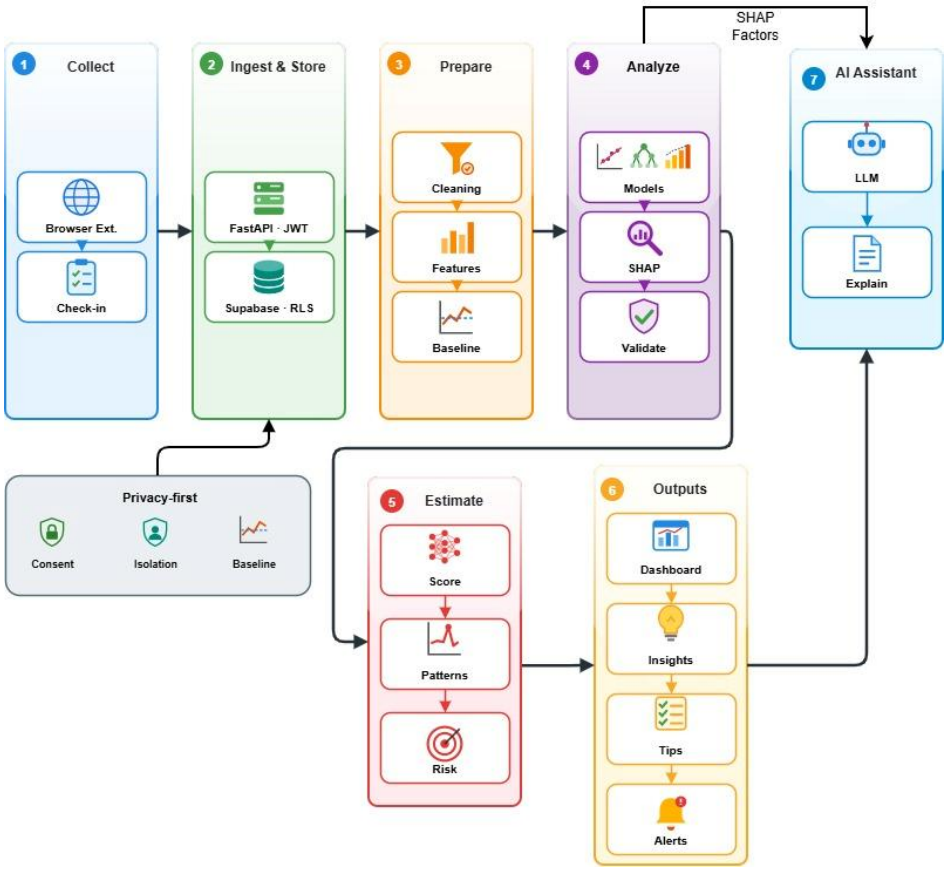


Fig. 1 Overall workflow of the SignalWell digital wellbeing platform. Verify that the inserted image is the original architecture diagram.

3.4 Database-Level Row-Level Security

PostgreSQL RLS policies are used to restrict access to records associated with the authenticated user. The policy pattern described in the implementation compares the authenticated identity with the record's user identifier:

```
ALTER TABLE daily_records ENABLE ROW LEVEL SECURITY;
CREATE POLICY user_isolation_policy ON daily_records
FOR ALL
USING (auth.uid() = user_id)
WITH CHECK (auth.uid() = user_id);
```

This is an illustrative policy from the manuscript. Confirm that the deployed schema, policy coverage, roles, and service credentials enforce the intended access restrictions before making a security claim.

Table 2 Core features described for SignalWell.

Feature	Type	Representation
Sleep duration	Continuous	Total hours of sleep per day
Screen time	Continuous	Aggregate daily minutes
Movement level	Continuous	Normalized daily activity measure
Subjective mood	Ordinal/categorical	Self-reported five-point scale
Stress response	Ordinal/categorical	Self-reported five-point scale
Anxiety self-assessment	Ordinal/categorical	Self-reported five-point scale
Personal baseline	Continuous	Moving average over seven preceding valid days
Risk indicator (target)	Continuous	Scale from 0.0 to 10.0

3.5 Runtime Model-Registry Verification

Before processing prediction requests, the backend checks whether the model registry is available. If the registry or required model artifacts are missing or invalid, the intended behaviour is to reject inference rather than return a prediction. This mechanism reduces the chance of inference proceeding with unavailable artifacts; it does not, by itself, prove model correctness or artifact integrity.

3.6 Ethics and Data Handling

The study reports that simulated data were used and that no clinical or participant data were collected. Any future study involving people should obtain the required ethics review and informed consent before data collection. Participants should be informed about the collected measures, purposes, storage, access, retention, and deletion. The platform is a research prototype, not a diagnostic tool.

3.7 Evaluation Metrics

Predictive performance was assessed using mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). Security testing used the access breach rate, defined as the proportion of tested unauthorized requests that were not blocked. These metrics address different aspects of the prototype and should not be conflated.

4 Experimental Setup

4.1 Hardware and Software

The manuscript reports Python 3.11/3.12, Node.js 20+, FastAPI with Uvicorn, Next.js 14/React, Supabase/PostgreSQL with RLS, and an analytics stack comprising scikit-learn, pandas, NumPy, and joblib. The reported compute host had an eight-core CPU and 16 GB RAM, without a GPU. Verify exact version numbers from environment files before final submission.

Table 3 Dataset and evaluation setup as reported in the manuscript.

Item	Reported value
Simulated daily records	2,500
User accounts represented	120
Training/testing split	80:20, time-based
Personal-baseline window	Seven preceding valid days
Automated pipeline tests passed	9 of 9 (100%)

4.2 Simulated Dataset

The paper reports 2,500 simulated daily records representing 120 user accounts. Sleep duration ranges from 3.5 to 11.5 hours and screen time from 45 to 720 minutes per day. The stated model split is an 80:20 time-based training/testing split, with earlier records used for training and later records for testing. These counts and ranges should be confirmed against the exact dataset and evaluation scripts. Results from simulated records may not generalize to real users.

4.3 Security Test Scenarios

Automated tests used a valid JWT for one user and attempted to read or modify records associated with another user. The manuscript reports more than 1,500 such requests and that all unauthorized requests in the tested scenarios were blocked. This result describes the tested cases only; it is not a guarantee that all possible security vulnerabilities or access paths have been eliminated.

5 Results and Discussion

5.1 Security Validation and Cross-Tenant Isolation

The manuscript reports that more than 1,500 unauthorized cross-account requests were blocked. The resulting observed access breach rate was 0%, with a 100% blocking rate in those tests. This supports the tested RLS isolation scenario, subject to the test coverage and deployed configuration.

5.2 Model-Registry Interception

When the registry or model weights were missing, or the registry schema was malformed, the system reportedly returned HTTP 503 rather than making a prediction. The manuscript reports detection in approximately 12 ms. The timing should be accompanied by a description of the measurement method and test environment if retained as a quantitative result.

5.3 Predictive Performance on the Risk-Indicator Task

Four approaches were compared on the simulated risk-indicator task. The reported gradient-boosting model (XGBoost) achieved the lowest MAE and RMSE and the

Table 4 Reported model-registry behaviour under fault conditions.

Registry state	Status	API response
Registry and model file present	Verified	200 OK with inference output
Registry file missing	Missing	503 Service Unavailable
Model weights missing	Fault	503 Service Unavailable
Malformed registry schema	Invalid	503 Service Unavailable

Table 5 Reported model performance on the simulated risk-indicator task.

Model	MAE	RMSE	R^2
Population static thresholds	2.14	2.89	0.28
Linear regression (baseline-relative)	1.08	1.42	0.61
Random forest regressor	0.64	0.88	0.82
Gradient boosting (XGBoost)	0.58	0.79	0.85

Table 6 Reported alert comparison on simulated data.

Metric	Fixed threshold	Personalized baseline
Precision	0.821	0.449
Recall	0.478	0.529
F1-score	0.604	0.486
False-alert rate	0.004	0.027
Detection delay (days)	0.74	0.23
Event detection rate	79.8%	92.7%
Alerts per user per week	0.16	0.33

highest R^2 among the compared approaches. These values describe performance on the reported simulated test set and should not be interpreted as clinical prediction accuracy.

5.4 Personalized-Baseline Alert Evaluation

The manuscript reports an alert comparison using simulated data from 500 users over 180 days. Fixed-threshold alerts achieved precision of 0.821, recall of 0.478, F1-score of 0.604, and false-alert rate of 0.004. Personalized-baseline alerts achieved precision of 0.449, recall of 0.529, F1-score of 0.486, and false-alert rate of 0.027. The reported event detection rate was 79.8% for fixed thresholds and 92.7% for personalized baselines; average detection delay was 0.74 and 0.23 days, respectively. The table indicates a trade-off: personalized alerts detected more simulated events sooner but had lower precision and a higher false-alert rate. The event-generation and metric-calculation procedures should be reported clearly to support reproducibility.

Table 7 Reported mean absolute SHAP values.

Feature	Mean absolute SHAP value
Sleep-duration deviation	0.42
Screen-time deviation	0.35
Stress-response deviation	0.28
Mood deviation	0.19
Movement deviation	0.12

5.5 Feature Importance

The manuscript reports a SHAP analysis in which sleep-duration deviation had the largest mean absolute SHAP value (0.42), followed by screen-time deviation (0.35), stress-response deviation (0.28), mood deviation (0.19), and movement deviation (0.12). These values indicate model attribution within the evaluated simulated task; they do not establish that the features cause changes in wellbeing.

5.6 Limitations and Safety

The prototype requires seven days of data to establish a personal baseline. Mood, stress, and anxiety are self-reported, and the model evaluation uses simulated rather than real-world records. Broad activity categories are used instead of fine-grained interaction logs, which supports data minimization but limits behavioural detail. Further work should evaluate the system with appropriate ethics oversight, representative data, pre-specified outcomes, and independent validation. SignalWell does not diagnose mental-health conditions or replace professional assessment.

6 Conclusion and Future Work

This paper presented SignalWell, a research prototype for studying changes in digital activity and self-reported wellbeing. The system combines personal-baseline comparison, database-level access controls, and model-registry checks. In the reported simulated evaluation, XGBoost achieved MAE 0.58, RMSE 0.79, and $R^2 = 0.85$. The security tests reported that unauthorized cross-account requests were blocked in the tested scenarios. Personalized-baseline alerts detected a greater proportion of simulated events with shorter reported delay, but also produced lower precision and a higher false-alert rate than fixed thresholds. These findings are preliminary and limited to simulated data. Future work should focus on reproducible evaluation, calibration, privacy and security review, and ethically approved real-world studies before any claims about practical or clinical effectiveness are made.

Declarations

Funding: No funding statement was provided in the source manuscript; complete this section accurately before submission.
Conflict of interest: The authors should declare any relevant competing interests.

Ethics approval: The source manuscript states that simulated data were used and no participant data were collected. Confirm the exact statement appropriate to the target journal.
Consent to participate: Not applicable to the reported simulated-data study, subject to confirmation by the authors.
Data availability: Add a statement identifying whether the simulated dataset and code will be shared, and where.
Author contributions: Add each author's verified contributions.
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